

Omarchy LMS Tier 3 Development Plan (v3)

Author: AI Engineering Lead & System Architect

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1. Executive Summary

This plan defines the engineering lifecycle and design specifications for an advanced Tier 3 Lyrion Music Server (LMS) frontend integration inside the Omarchy tiling ecosystem. By shifting away from memory-heavy browser instances, this plugin bridges the gap between terminal efficiency, unified system searching, keyboard-centric hotkeys, and AI agent execution. The architecture relies heavily on a decoupled single-binary engine communicating over raw TCP and JSON-RPC protocols.

2. Programming Language Architectural Tradeoffs

Choosing the language for the core CLI execution binary requires weighing execution speed, developer velocity, type-safety, and ecosystem support for Terminal User Interfaces (TUIs).

Metric	Go Language Architecture	Rust Language Architecture
Development Velocity	Very High. Standard library features native JSON, HTTP, and TCP modules out of the box.	Medium. Rigid borrow checker and memory safety paradigms prolong initial prototyping.
TUI Ecosystem Support	Excellent. The Bubble Tea (Charm) framework offers high-performance reactive paradigm frameworks.	Industry Standard. Ratatui delivers exceptional precision but requires significant state-management boilerplate.
Memory & Compile footprint	~5MB to 10MB idle binary footprint due to bundled runtime/garbage collector.	~1MB to 3MB total compile size with strict, dependency-free binary compilation.
Concurrency Architecture	Goroutines provide first-class channels for lightning-fast concurrent player polls.	Async/Await requires careful crate synchronization (e.g., Tokio).

3. Advanced Configurable Installer Matrix

The out-of-box installer maps configuration files interactively to eliminate keyboard layout limitations and cross-application keybinding collisions.

Pre-Check Automations:

- **Keyboard Layout Validation:** Scans localectl and XKB configurations to preempt symbol shifts on non-US QWERTY setups (e.g., AZERTY/QWERTZ).
- **Duplicate Binding Verification:** Employs regex strings across hyprland.conf to block shortcut collisions against core desktop window navigation bindings.

Interactive Prompting Matrix:

- **Prompt 1 (Physical Layout Layouts):** Selection between Hardware Media keys (XF86Audio), Minimalist Arrow Modifiers (SUPER + Arrows), or Home-Row Developer brackets (SUPER + [/]).
- **Prompt 2 (Contextual Priority Policies):** Resolves active playback overrides (e.g., background LMS skipping vs active foreground YouTube video playback) utilizing strict shift modifier blocks or real-time query evaluations.

4. Shell Interface & High-Density TUI Layout

The interface splits playback responsibilities into two distinct visuals:

- **Quickshell Top-Bar Widget:** A lightweight, asynchronously refreshed JavaScript widget rendering track timelines alongside clickable fallback micro-controls.
- **Unified Menu Hooks:** Feeds local database search matrices into wofi/rofi interfaces via customized text formatting.
- **Full Feature TUI Application:** Launches an interactive terminal dashboard featuring active room-synchronization control matrices and a JSON-Schema bridge for automated AI system agents (omp/ori).